Academic case study | Sophia Learning — Introduction to Networking
GreenLeaf Seattle Network Vendor Evaluation and Recommendation

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Executive Summary

For GreenLeaf's new Seattle office, I recommend Vendor B, Pacific Northwest Networking. The Seattle location needs a network that can support the company's current operations while also providing enough security, reliability, and room for future growth. After comparing all three proposals, Vendor B received the highest weighted score of 4.20 out of 5.00 and provided the best overall balance between technical capabilities and cost. Its proposal includes dual internet connections, stronger firewall protection, redundant core switching, Wi-Fi 6, and support for future growth while still staying within GreenLeaf's five-year budget. Vendor A is less expensive but has more single points of failure and limited scalability, while Vendor C provides the strongest technical solution but is significantly over budget. For these reasons, I think Vendor B is the most practical and well-balanced choice for the Seattle office.

Analysis Summary

The requirements that had the biggest influence on my evaluation were reliability, security, future growth, and GreenLeaf's $275,000 five-year budget. Seattle needs a network that can support 25% annual growth without major changes, with infrastructure capable of supporting up to 100 users, while maintaining 99.9% availability. Because of these requirements, I paid close attention to each vendor's WAN design, switching capacity, wireless network, security controls, and redundancy. I also considered whether the additional features offered by each vendor were actually worth the cost. Vendor A had the lowest five-year cost, but its single ISP, single router, and single core switch created

more risk and made the network less prepared for future growth. Vendor C had the strongest technical design and the most redundancy, but its five-year cost was far beyond GreenLeaf's budget and its proposed implementation timeline did not meet the project deadline. Vendor B provided a better balance by offering dual internet connections, redundant core switching, a dedicated next-generation firewall, Wi-Fi 6, and room for future expansion while remaining within the five-year budget. This was also reflected in the scoring matrix, where Vendor B received the highest overall score of 4.20, compared with 3.80 for Vendor C and 3.40 for Vendor A.

Vendor Comparison and Scoring Matrix

Table 1. Vendor Weighted Scoring Matrix

Evaluation Category	Weight	Vendor A	Vendor B	Vendor C
Technical Solution	35%	3/5	4/5	5/5
Cost	25%	5/5	4/5	1/5
Reliability & Redundancy	20%	2/5	4/5	5/5
Vendor Qualifications	10%	3/5	5/5	5/5
Implementation Plan	10%	4/5	5/5	3/5
Weighted Total	100%	3.40/5	**4.20/5**	3.80/5

The scoring matrix shows that Vendor B received the highest overall score at 4.20 out of 5.00. Vendor A received the highest cost score because it was the most affordable option, but its single ISP, router, and core switch lowered its reliability score. Vendor C received the highest technical and reliability scores because it offered the strongest network design and the most redundancy, but its cost was significantly over GreenLeaf's budget, resulting in a much lower cost score. Vendor B consistently scored well across all five categories because it provided strong technical capabilities, better redundancy, experienced vendor support, and a realistic implementation plan while still remaining within GreenLeaf's five-year budget. This balance is what gave Vendor B the highest weighted score overall.

Strengths of Your Recommended Vendor

One of Vendor B's biggest strengths is that its proposal gives GreenLeaf a good balance between reliability, security, performance, and future growth without going beyond GreenLeaf's $275,000 five-year budget limit. The dual internet connections with automatic failover provide more reliable access to the internet and Portland resources, while the stacked core switches add redundancy within the Seattle network. Vendor B also includes a dedicated FortiGate next-generation firewall with features such as intrusion prevention, web filtering, and application control, which provides stronger security than relying only on the router for firewall protection. The four Wi-Fi 6 access points provide newer wireless technology and enough capacity to support the expected increase in employees and wireless devices. The design also includes additional switch capacity and expansion VLANs, which will make it easier for GreenLeaf to grow the Seattle office without making major changes to the network. Another important strength is that Vendor B integrates with GreenLeaf's existing Portland systems, including PRTG monitoring, syslog, RADIUS authentication, and the site-to-site VPN. Overall, these features make the proposal strong enough to support Seattle's future needs without adding the cost and complexity of an enterprise-level solution that the office may not actually need.

Limitations and Mitigations

Even though Vendor B is the strongest overall option, there are still a few limitations that GreenLeaf should consider. One concern is that the design uses a single Cisco ISR 4331 router and a single FortiGate 100F firewall, which means these devices could still become single points of failure even though the proposal includes dual internet connections and redundant core switches. GreenLeaf could reduce this risk in the future by adding a backup router or a second firewall in a high-availability configuration if the Seattle office continues to grow. Another limitation is that the FortiGate firewall is not Cisco equipment, while GreenLeaf generally prefers Cisco devices. However, Vendor B is a Fortinet Expert Partner and provides 24/7 support, which should help reduce concerns about managing a different platform. The proposal's five-year TCO of $262,660 is relatively close to GreenLeaf's $275,000 budget limit, so future upgrades and licensing costs should

be monitored carefully. Overall, these limitations are manageable and do not outweigh the reliability, security, and scalability benefits that Vendor B provides.

Implementation Considerations

If GreenLeaf selects Vendor B, the company will need to carefully plan the implementation so the Seattle network is ready before the required deadline. Vendor B estimates an eight-week deployment, which includes planning and procurement, cabling and infrastructure work, installation of the router, firewall, and switches, followed by VPN, wireless, VoIP, and QoS configuration. The final stage includes user acceptance testing, employee training, documentation, and a period of parallel operation before the project is handed over to GreenLeaf. Because Vendor B's eight-week implementation schedule leaves limited room before GreenLeaf's July 15 hard deadline, equipment availability and installation progress should be monitored closely to prevent delays. GreenLeaf's IT staff will also need to coordinate with Vendor B during the Portland integration, especially for the site-to-site VPN, RADIUS authentication, PRTG monitoring, and syslog services. Testing the failover connections, security controls, wireless coverage, and access to Portland resources before the final cutover will help identify problems early. Vendor B also includes training and 24/7 support for critical issues, which should make the transition easier and help GreenLeaf manage the network successfully after implementation.

Alternatives Considered

Vendor A and Vendor C were both considered as alternatives to Vendor B. Vendor C came the closest to being selected, with a weighted score of 3.80 out of 5.00 compared with Vendor B's 4.20. Vendor C offered the strongest technical solution and the highest level of reliability, including redundant routers, firewalls, core switches, and internet connections. However, its five-year cost of $609,300 is significantly higher than GreenLeaf's $275,000 maximum budget, and its 12-week implementation timeline would also go beyond the July 15 hard deadline. Vendor A was the most affordable option and received the highest cost score, but its single ISP, single router, and single core switch created more single points of failure and made the design less suitable for GreenLeaf's future growth. When both the proposal features and final scoring results are considered,

Vendor B provides a better balance of cost, reliability, security, and scalability than either alternative, making it the best overall choice for the Seattle office.